Introduction

This book is based upon a combination of firmly established physical principles and highly imaginative speculation, in which the author attempts to reach beyond what is solidly known at present and project into a part of the physical world that has no known counterpart in our everyday life on Earth. His goal is, among other things, to examine both the exterior and interior of a black hole—a stellar body so massive and concentrated that its gravitational field prevents material particles and light from escaping in ways which are common to a star such as our own Sun. The descriptions given of events that would be experienced if an observer were to approach such a black hole from outside are based upon predictions of the general theory of relativity in a "strong-gravity" realm where it has never yet been tested. The speculations which go beyond that and deal with the region inside what is termed the black hole's "horizon" are based on a special form of courage, indeed of bravado, which Thorne and his international associates have in abundance and share with much pleasure. One is reminded of the quip made by a distinguished physicist, "Cosmologists are usually wrong but seldom in doubt." One should read the book with two goals: to learn some hard facts with regard to strange but real features of our physical Universe, and to enjoy informed speculation about what may lie beyond what we know with reasonable certainty.

As a preface to the work, it should be said that Einstein's general theory of relativity, one of the greatest creations of speculative science, was formulated just over three-quarters of a century ago. Its triumphs in the early 1920s in providing an explanation of the deviations of the motion of the planet Mercury from the predictions of the Newtonian theory of gravitation, and later an explanation of the redshift of distant nebulas discovered by Hubble and his colleagues at Mount Wilson Observatory,